

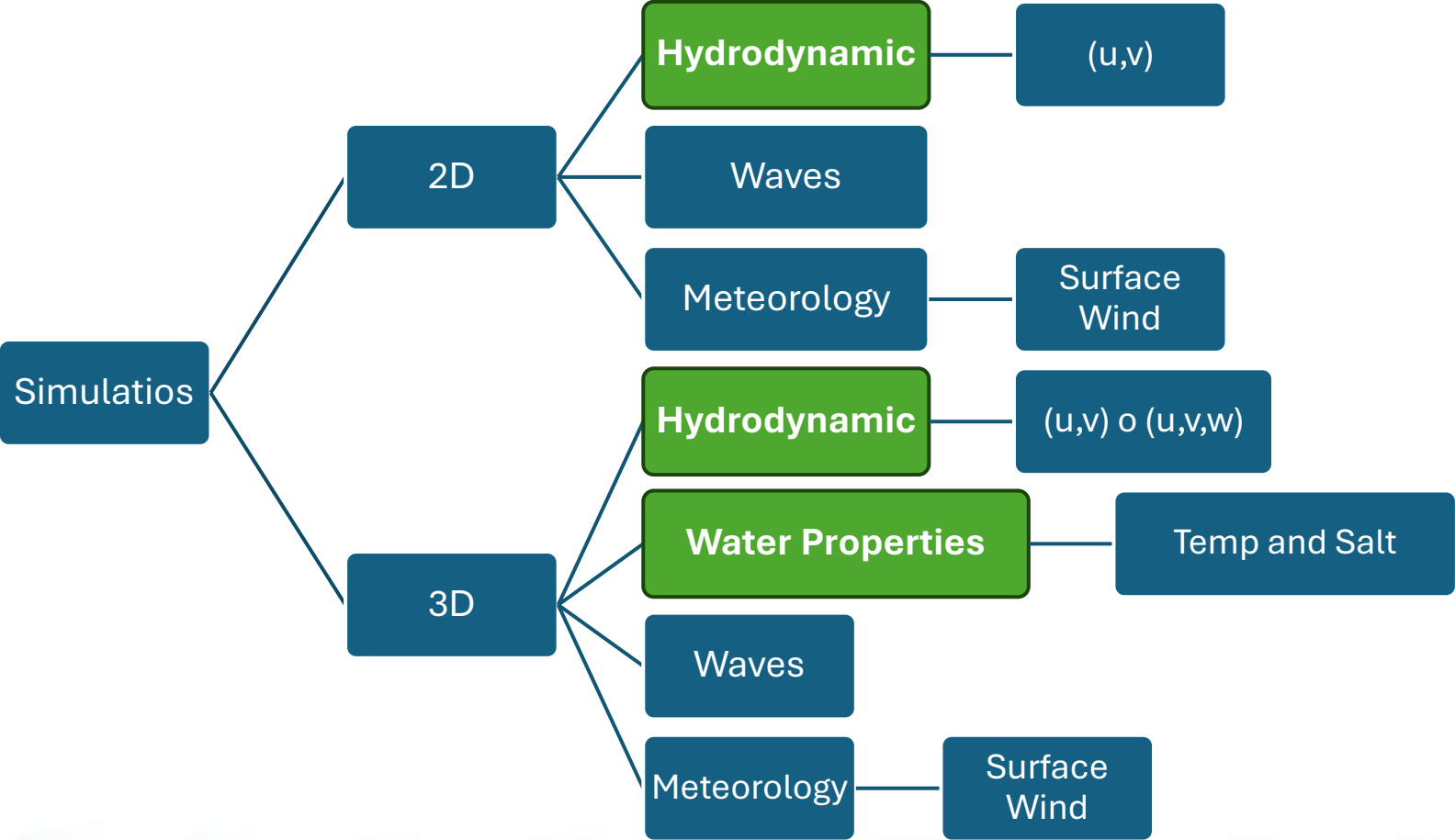


Activity 2: Download oceanographic data and prepare it for MOHID use

Sara Cloux Gonzalez¹

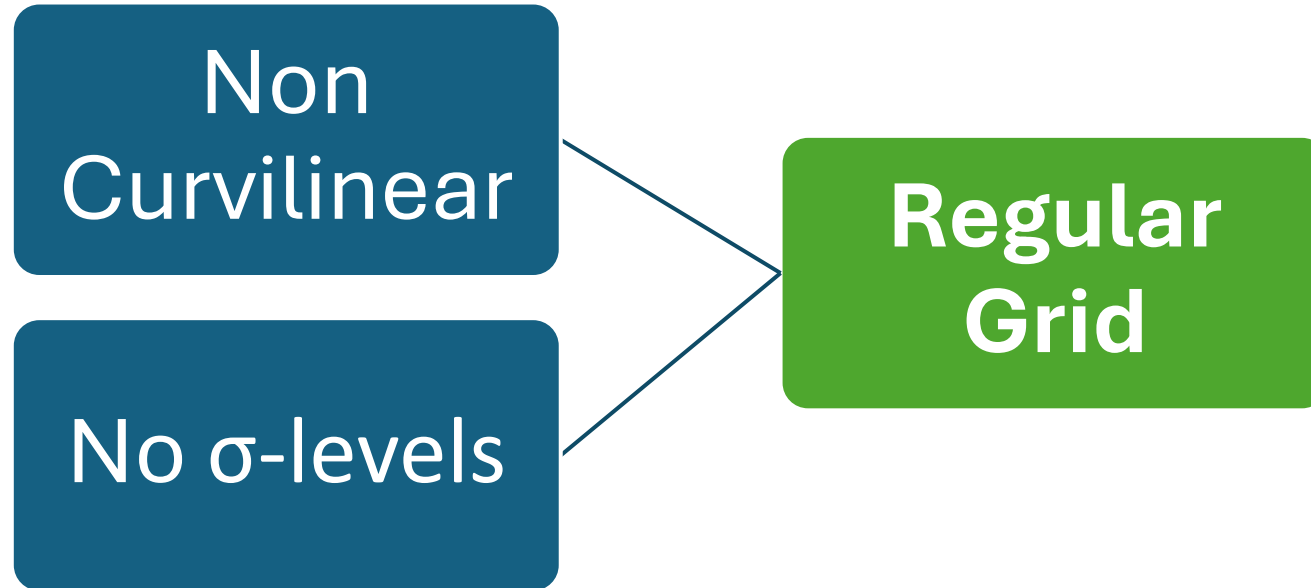
¹ Marine Institute, Foras na Mara, Galway, Ireland..

Input files for MOHID



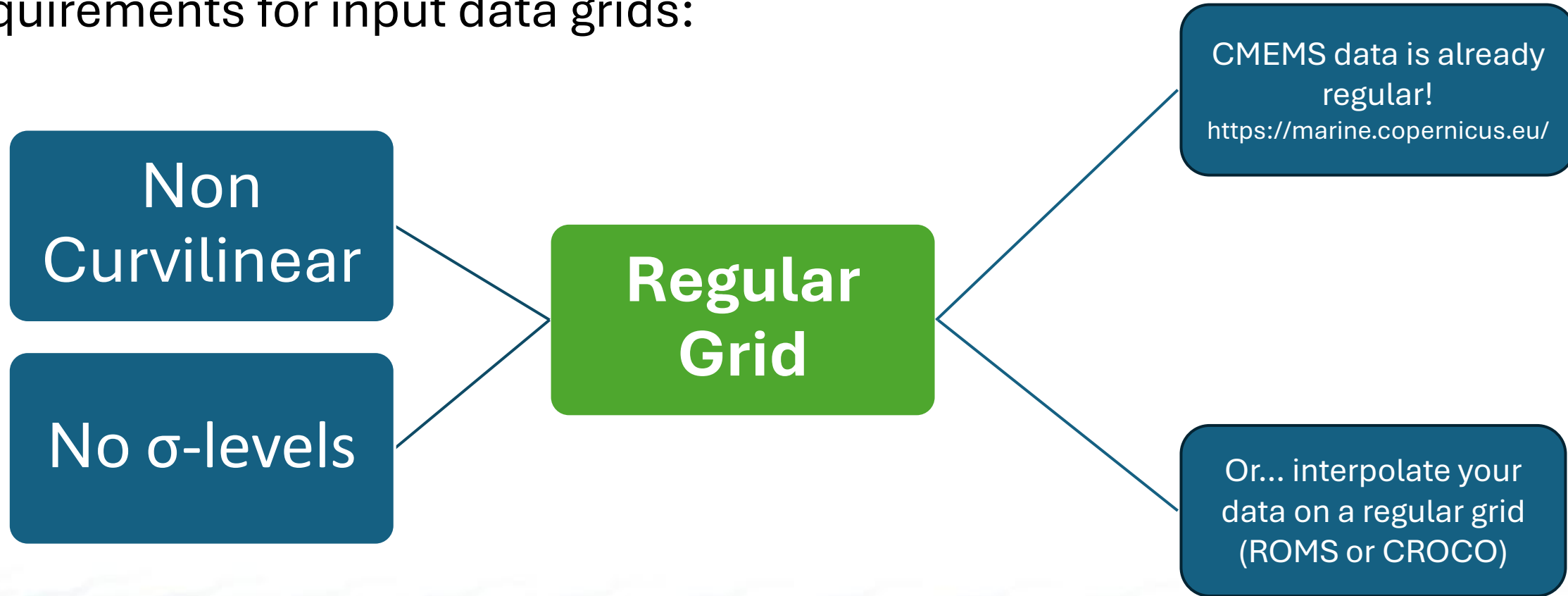
Input files for MOHID

Requirements for input data grids:



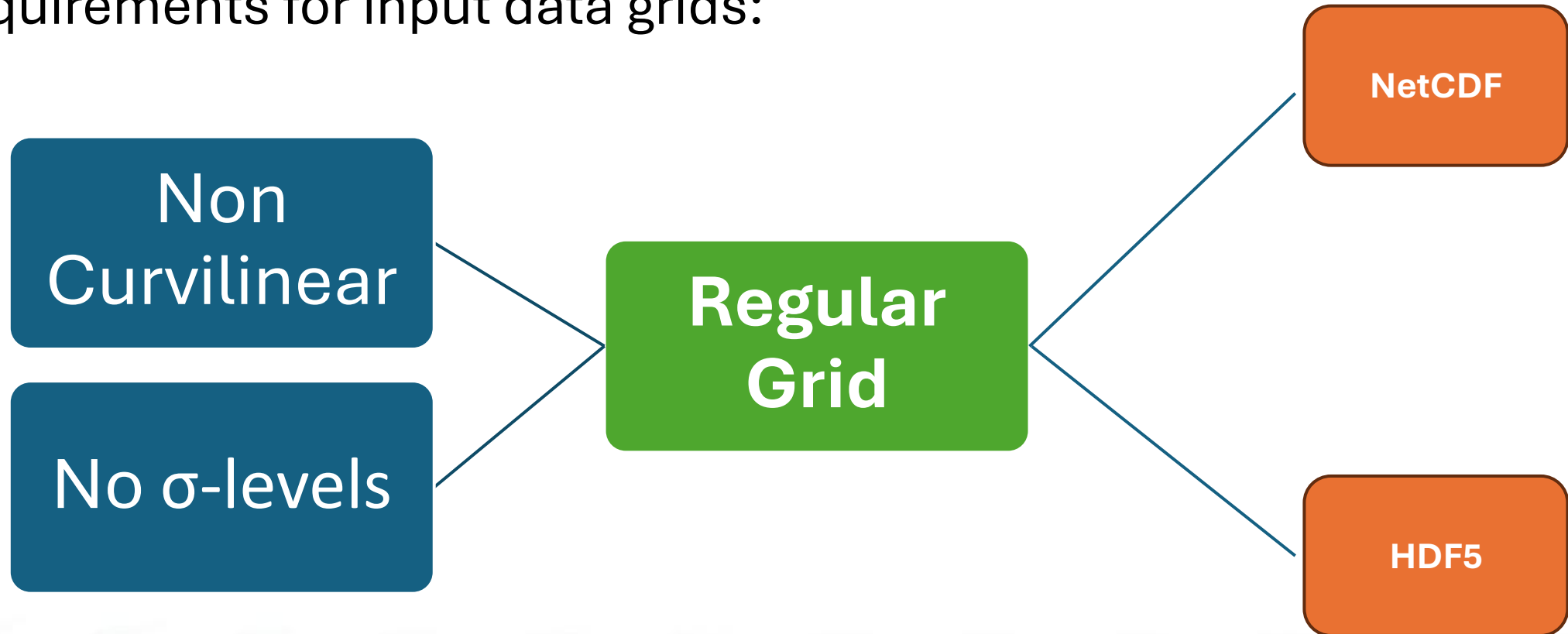
Input files for MOHID

Requirements for input data grids:



Input files for MOHID

Requirements for input data grids:



*How download CMEMS data

Copernicus Marine Data Store

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Type...

English



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PROGRAMME OF
THE EUROPEAN UNION



Implemented by
**MERCATOR
OCEAN**
INTERNATIONAL

X close

Access data

Home > Access data

Explore terabytes of ocean data at all different user levels

DATA

OCEAN PRODUCTS

Ocean product catalogue, to download or visualize data across nearly 15 variables, including hindcast, current and forecast data

TRENDS

OCEAN CLIMATE TRENDS

Monitoring the health of the ocean.
[Ocean Monitoring Indicators](#)
[Ocean Climate Portal](#)

EXPERTISE

OCEAN STATE REPORT

Extensive annual analysis on the state of the ocean over nearly 20 years and severe/notable annual events

TRACK

OCEAN IN SITU

Track in situ technology deployed in the ocean through our In Situ TAC Dashboard

*How download CMEMS data

Python API: copernicusmarine

```
import copernicusmarine as cop

min_lon, max_lon = -19, 0.5
min_lat, max_lat = 30, 55.5
min_dep, max_dep = 0.5, 565.29

start = "2023-03-01T00:00:00"
end = "2023-08-31T23:59:59"

10
11 out_dir = '/home/matlab/saracloux/data/copernicus/hydro_ibi/'
12
13 u_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'
14 v_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'
15 w_url = 'cmems_mod_ibi_phy_my-wcur_0.083deg_P1D-m'
16 temp_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'
17 salt_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'
18
19 bathy_url = 'cmems_mod_ibi_wav_my_0.027deg-static'
20 url_list = [bathy_url]
21 var_codes = ['deptho']
22 var_names = ['deptho']
23
24
25 for i in range(len(var_codes)):
26     var = var_codes[i]
27     var_name = var_names[i]
28     var_url = url_list[i]
29     print(f'Downloading {var_name} ...')
30     dataset = cop.subset(
31         dataset_id = var_url,
32         variables = [var],
33         start_datetime = start,
34         end_datetime = end,
35         minimum_longitude = min_lon,
36         maximum_longitude = max_lon,
37         minimum_latitude = min_lat,
38         maximum_latitude = max_lat,
39         # minimum_depth = min_dep,
40         # maximum_depth = max_dep,
41         output_filename = f'{var_name}.nc',
42         output_directory = out_dir
43     )
44     print(f'{var_name} Downloaded succesfully')
45     print()
```

*How download CMEMS data

```
import copernicusmarine as cop

min_lon, max_lon = -19, 0.5
min_lat, max_lat = 30, 55.5
min_dep, max_dep = 0.5, 565.29

start = "2023-03-01T00:00:00"
end = "2023-08-31T23:59:59"

out_dir = '/home/matlab/saracloux/data/copernicus/hydro_ibi/'

u_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'
v_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'
w_url = 'cmems_mod_ibi_phy_my-wcur_0.083deg-P1D-m'
temp_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'
salt_url = 'cmems_mod_ibi_phy_my_0.083deg-3D_P1D-m'

bathy_url = 'cmems_mod_ibi_wav_my_0.027deg-static'
url_list = [bathy_url]
var_codes = ['deptho']
var_names = ['deptho']

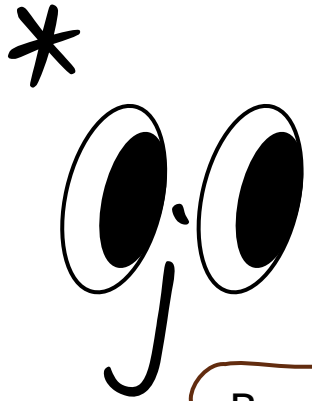
for i in range(len(var_codes)):
    var = var_codes[i]
    var_name = var_names[i]
    var_url = url_list[i]
    print(f'Downloading {var_name} ...')
    dataset = cop.subset(
        dataset_id = var_url,
        variables = [var],
        start_datetime = start,
        end_datetime = end,
        minimum_longitude = min_lon,
        maximum_longitude = max_lon,
        minimum_latitude = min_lat,
        maximum_latitude = max_lat,
        # minimum_depth = min_dep,
        # maximum_depth = max_dep,
        output_filename = f'{var_name}.nc',
        output_directory = out_dir
    )
    print(f'{var_name} Downloaded succesfully')
    print()
```



This data could need some preprocessing in terms of **time resolution** and **variable names**.

Input files for MOHID

ncNamesLibrary.xml

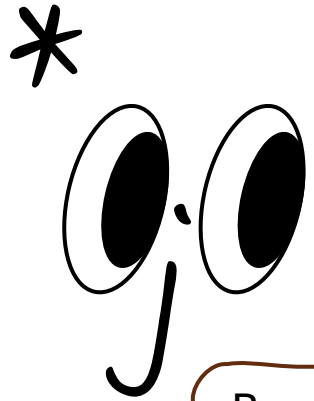


Be careful with variable names and dimension names

```
ncNamesLibrary.xml x
1 <?xml version="1.0" encoding="UTF-8" ?>
2 <naming>
3   <variables>
4     <eastward_sea_water_velocity name="u">
5       <variant name="u" comment="used in MOHID" />
6       <variant name="velocity U" comment="used in MOHID" />
7       <variant name="uu" />
8       <variant name="U" />
9       <variant name="uo" comment="used in CEMS" />
10      <variant name="u_east" comment="used by SINTEF" />
11    </eastward_sea_water_velocity>
12    <northward_sea_water_velocity name="v">
13      <variant name="v" comment="used in MOHID" />
14      <variant name="velocity V" comment="used in MOHID" />
15      <variant name="vv" />
16      <variant name="V" />
17      <variant name="vo" comment="used in CEMS" />
18      <variant name="v_north" comment="used by SINTEF" />
19    </northward_sea_water_velocity>
20    <upward_sea_water_velocity name="w">
21      <variant name="w" comment="used in MOHID and CEMS" />
22      <variant name="W" />
23      <variant name="velocity W" />
24    </upward_sea_water_velocity>
25    <sea_water_temperature name="temp">
26      <variant name="temp" comment="used in MOHID and CEMS" />
27      <variant name="Temp" />
28      <variant name="temperature" />
29      <variant name="Temperature" />
30    </sea_water_temperature>
31    <sea_water_salinity name="salt">
32      <variant name="salt" comment="used in MOHID and CEMS" />
33      <variant name="salinity" />
34      <variant name="Salt" />
35      <variant name="Salinity" />
36    </sea_water_salinity>
37    <bathymetry name="bathymetry">
38      <variant name="bathymetry" comment="used in MOHID and CEMS" />
39      <variant name="Bathymetry" comment="used in MOHID and CEMS" />
40      <variant name="bathym" />
41    </bathymetry>
```

Input files for MOHID

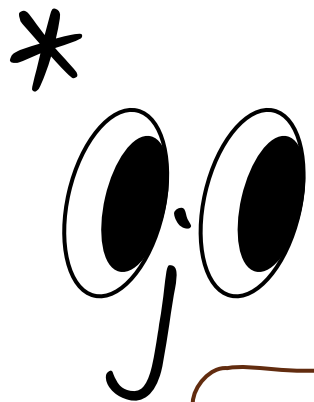
ncNamesLibrary.xml



Be careful with variable names and dimension names

```
</variables>
<dimensions>
  <longitude name="lon">
    <variant name="lon" />
    <variant name="Lon" />
    <variant name="LON" />
    <variant name="longitude" />
    <variant name="Longitude" />
    <variant name="LONGITUDE" />
  </longitude>
  <latitude name="lat">
    <variant name="lat" />
    <variant name="Lat" />
    <variant name="LAT" />
    <variant name="latitude" />
    <variant name="Latitude" />
    <variant name="LATITUDE" />
  </latitude>
  <vertical name="level">
    <variant name="zc" />
    <variant name="depth" />
    <variant name="Depth" />
    <variant name="DEPTH" />
    <variant name="level" />
    <variant name="Level" />
    <variant name="VerticalZ" />
    <variant name="Vertical" />
  </vertical>
  <time name="time">
    <variant name="time" />
    <variant name="Time" />
    <variant name="TIME" />
  </time>
</dimensions>
</naming>
```

Input files for MOHID



Be careful time
resolution of input files

Input files for MOHID

```
1 <?xml version="1.0" encoding="UTF-8" ?>
2 <case>
3   <execution>
4     <parameters>
5       <parameter key="Start" value="2024 02 02 00 00 00" comment="Date of initial instant" units_comment="space delimited ISO 8601 format up to seconds" />
6       <parameter key="End" value="2024 02 02 22 00 00" comment="Date of final instant" units_comment="ISO format" />
7       <parameter key="BaseDateTime" value="1950 01 01 00 00 00" comment="Base Date for time stamping purposes. Optional, default = 1950/1/1" units_comment="ISO format" />
8       <parameter key="Integrator" value="1" comment="Integration Algorithm 1:Euler, 2:Multi-Step Euler, 3:RK4 (default=1)" />
9       <parameter key="Threads" value="24" comment="Computation threads for shared memory computation (default=auto)" />
10      <parameter key="BufferSize" value="86400" comment="Optional parameter. Controls input frequency" units_comment="seconds" />
11    </parameters>
12    <parameter key="OutputFormat" value="2" comment="Output file format. NetCDF=1; VTK=2 (default=2)" />
13  </execution>
14  <variableNaming>
15    <file name="ncNamesLibrary.xml"/>
16  </variableNaming>
17  <outputFields>
18    <file name="data/outputFields.xml"/>
19  </outputFields>
20 </execution>
21 <caseDefinitions>
22   <inputData>
23     <inputDataDir name="nc_fields_2D/hydrodynamic" type="hydrodynamic"/>
24     <inputDataDir name="nc_fields_2D/waterProperties" type="waterProperties"/>
25     <inputDataDir name="nc_fields_2D/winds" type="meteorology"/>
26   </inputData>
27   <simulation>
28     <resolution dp="300" units_comment="metres (m)"/>
29     <timestep dt="900.0" units_comment="seconds (s)"/>
30     <BoundingBoxMin x="-9.10" y="42.39" z="-140" units_comment="(deg,deg,m)"/>
31     <BoundingBoxMax x="-8.72" y="42.68" z="3" units_comment="(deg,deg,m)"/>
32     <VerticalVelMethod value="1" comment="1:From velocity fields, 2:Divergence based, 3:Disabled. Default = 1" />

```

```
<parameter key="BufferSize" value="86400" comment="Optional parameter. Controls input frequency" units_comment="seconds" />
```

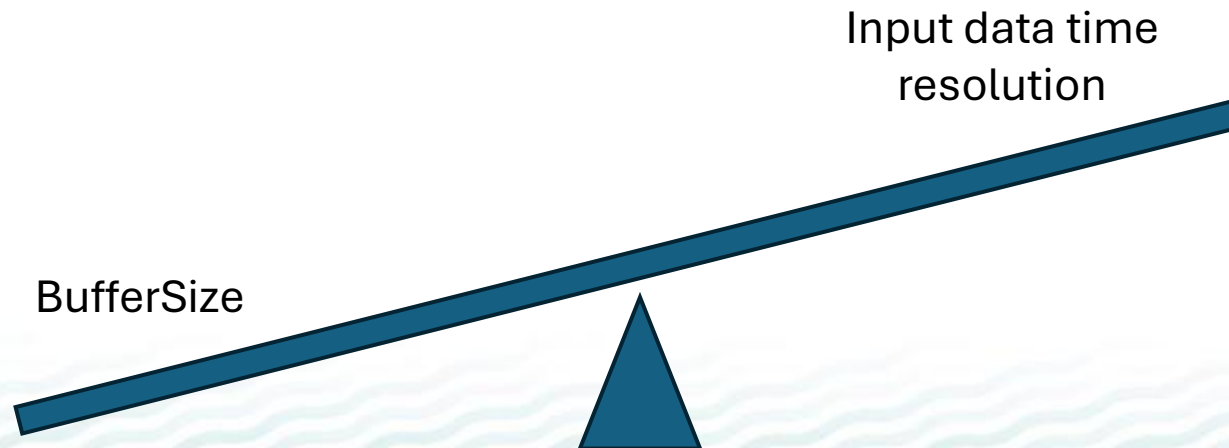
$86400 \text{ sec} \equiv 1 \text{ day of data}$

Input files for MOHID

```
<parameter key="BufferSize" value="86400" comment="Optional parameter. Controls input frequency" units_comment="seconds" />
```

$86400 \text{ sec} \equiv 1 \text{ day of data}$

1 input file of 3 hours $\rightarrow$ loading 8 files in memory



Input files for MOHID

```
1 <?xml version="1.0" encoding="UTF-8" ?>
2 <case>
3   <execution>
4     <parameters>
5       <parameter key="Start" value="2024 02 02 00 00 00" comment="Date of initial instant" units_comment="space delimited ISO 8601 format up to seconds" />
6       <parameter key="End" value="2024 02 02 22 00 00" comment="Date of final instant" units_comment="ISO format" />
7       <parameter key="BaseDateTime" value="1950 01 01 00 00 00" comment="Base Date for time stamping purposes. Optional, default = 1950/1/1" units_comment="ISO format" />
8       <parameter key="Integrator" value="1" comment="Integration Algorithm 1:Euler, 2:Multi-Step Euler, 3:RK4 (default=1)" />
9       <parameter key="Threads" value="24" comment="Computation threads for shared memory computation (default=auto)" />
10      <parameter key="BufferSize" value="86400" comment="Optional parameter. Controls input frequency" units_comment="seconds" />
11      <parameter key="OutputWriteTime" value="3600" comment="Time out data (1/Hz)" units_comment="seconds" />
12      <parameter key="OutputFormat" value="2" comment="Output file format. NetCDF=1; VTK=2 (default=2)" />
13    </parameters>
14    <variableNaming>
15      <file name="ncNamesLibrary.xml"/>
16    </variableNaming>
17    <outputFields>
18      <file name="data/outputFields.xml"/>
19    </outputFields>
20  </execution>
21  <inputData>
22    <inputDataDir name="nc_fields_2D/hydrodynamic" type="hydrodynamic"/>
23    <inputDataDir name="nc_fields_2D/waterProperties" type="waterProperties"/>
24    <inputDataDir name="nc_fields_2D/winds" type="meteorology"/>
25  </inputData>
26  <simulation>
27    <resolution dp="300" units_comment="metres (m)"/>
28    <timestep dt="900.0" units_comment="seconds (s)"/>
29    <boundingBoxMin x="-9.10" y="42.39" z="-140" units_comment="(deg,deg,m)"/>
30    <boundingBoxMax x="-8.72" y="42.68" z="3" units_comment="(deg,deg,m)"/>
31    <VerticalVelMethod value="1" comment="1:From velocity fields, 2:Divergence based, 3:Disabled. Default = 1" />
32  </simulation>
</case>
```

Before start, MOHID-Preprocessing reads **ALL** the files in the folder, no matter how much it needs for the simulation.